



DEPARTMENT OF ELECTRICAL & ELECTRONICS ENGINEERING
EFB 1054 DIGITAL ELECTRONICS
MAY 2026

ASSIGNMENT II

Course Outcome (CO 2): Use Boolean algebra and Karnaugh Maps to minimize Boolean expressions for the design of digital logic circuits.

1. For given function, $M(x, y, z) = \bar{x}y\bar{z} + \bar{x}yz + x\bar{y}\bar{z} + x\bar{y}z + xyz$.

- a) Construct truth table.
- b) Simplify the Boolean expression using Karnaugh map.

[10 marks]

2. You are tasked with designing a voting system using combinational logic. The system should accept votes from three sources: V1, V2, and V3, where each vote can be either '1' (in favor) or '0' (against). The system should output a majority decision (M), meaning that the output should be '1' if two or more of the votes are in favor, and '0' otherwise.

- a) Develop the truth table for the majority decision (M) based on the possible vote combinations.
- b) Simplify the Boolean expression using Karnaugh maps or Boolean algebra.
- c) Design the combinational logic circuit using the identified logic gates (AND, OR, NOT, etc.) to implement the majority voting system.
- d) Explain how this system could be extended to handle additional voters (e.g., four or five voters) and how the logic would change.

[10 marks]